

THE EMERGENT BELT

涌现之带

Jing-Zhang Hyper Line

Centennial Jing-Zhang AI Belt · open-call submission

KUN-SAL Spatial Quant Lab · Xing Xinhua

$\lambda_c=0.42$

connection

$H=1.556$

health

$D=1.746$

fractal

25,476

POIs

+15.9%

job reach

44亿

investment

Conceptual: provisional boundary dashed only; statutory indicators pending

01

Diagnosis: what blocks this belt

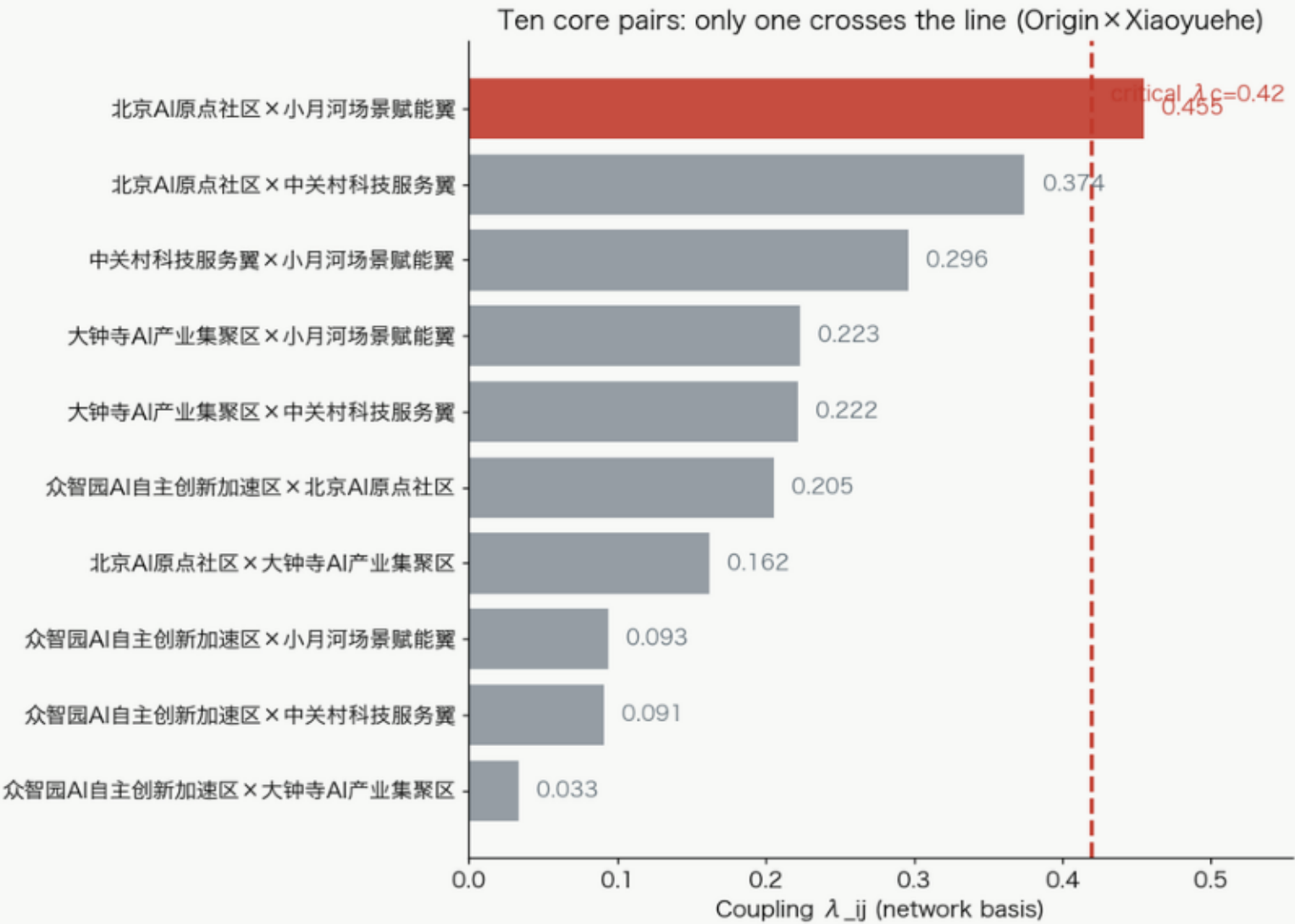
Five scientists' consultation · Space syntax · LUTI (jobs-housing dynamics) · POI dynamics · Boundary correction

① Spatial interaction λ: stuck subcritical

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Fig C1 · Wilson spatial-interaction λ calibration (status quo)

$\lambda_{ij} = \exp(-d_{ij}/2500m)$; $\lambda > \lambda_c=0.42$ = reachable within the 10-min innovation circle (OSM network)



Supercritical share · euclidean

Supercritical share · network

5-core global connectivity

50.00%

29.17%

No (severed)

Diagnosis: the five cores are not connected under the 10-min coupling — a healthy skeleton severed by seven roads (λ is cut one of the diagnosis ring).

- Only one of ten core pairs crosses the line (Origin × Xiaoyuehe 0.455)
- Zhongzhiyuan×Dazhongsi λ=0.033: adjacent on the map, unreachable in ten minutes
- Not without vitality — vitality chopped apart by seven roads
- The leverage point is the seven gaps; no even spreading

Basis: $\lambda=\exp(-d/2500m)$ proxy; recalculation on official data



② POI check-up: demand recognized by 25,476 points

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▪ 25,476 POIs across the belt (2026-08-15 snapshot)

Fig C7 · Amap POI dynamics: the invisible engine of the city

Fig C7a · 200m-grid POI density field (25476 pts, 12 categories)

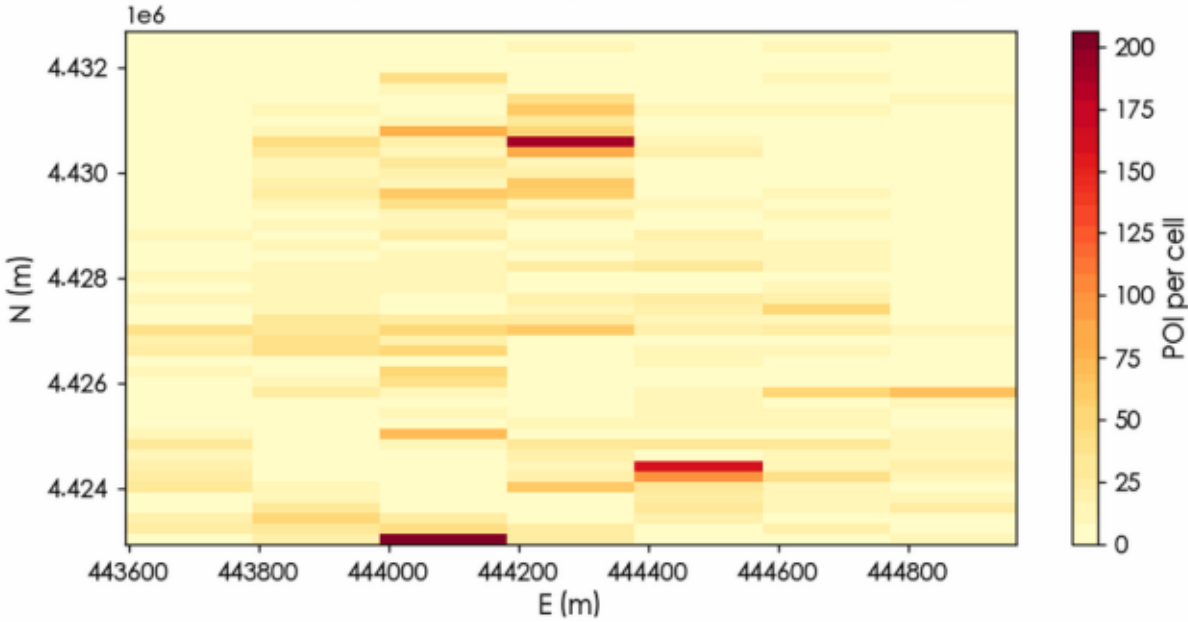
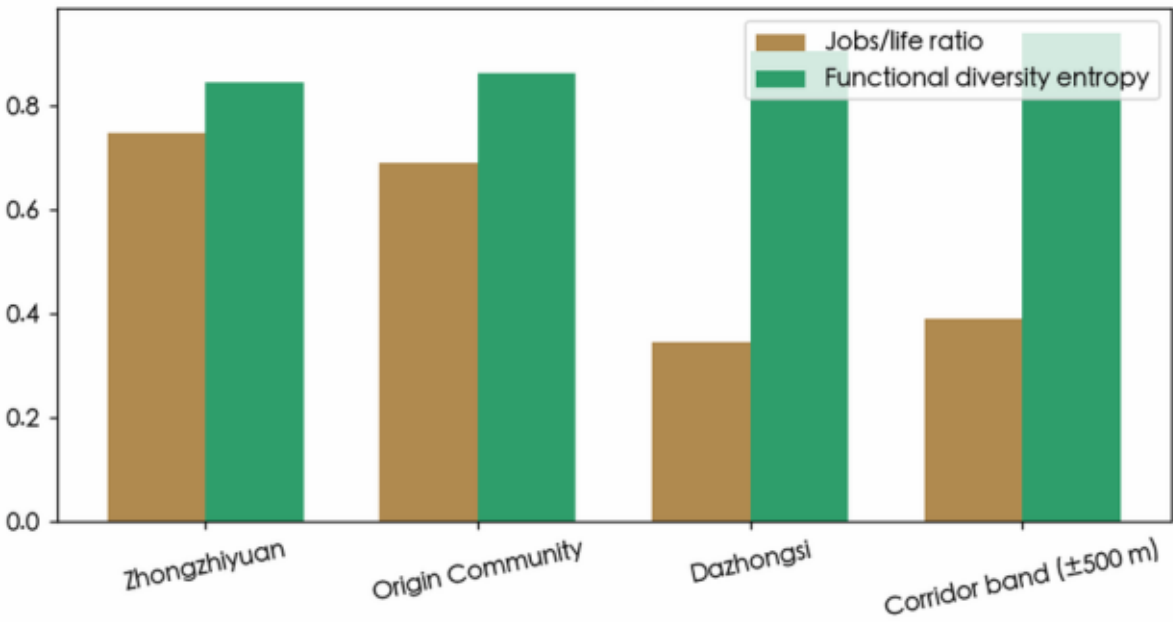


Fig C7b · Functional check-up of three areas + corridor



▪ Life services over-supplied; innovation content under-supplied

▪ Finance -63.0% / research -45.9% / Origin medical 2.88/km², the lowest

▪ Dazhongsi 511/km², the densest; jobs/life 0.344: life dense, work thin

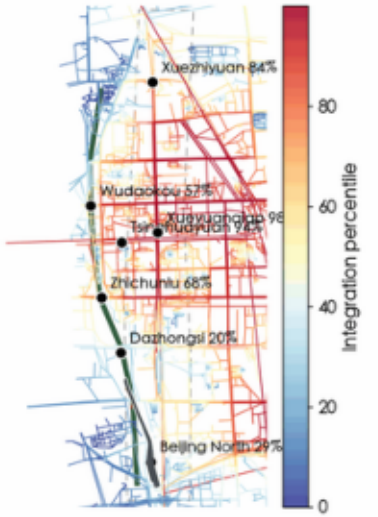
Data: Amap aggregate indicators; no raw points published

③ Space syntax: configuration shapes movement

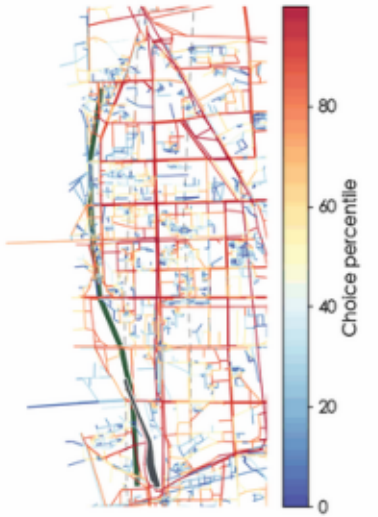
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Space syntax plan analysis: configuration shapes movement (5,720 nodes, computed)

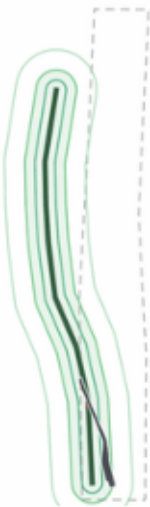
Map A · Integration: red = structural core



Map B · Choice: red = through-movement spine



Map C · 200/400/800m walking rings: the hardest 400m on the belt



▪ Xueyuanqiao 97.5%: the belt's mouth at the N4th Ring × Xueyuanlu

▪ Tsinghuayuan 93.5%: a 1909 siting still the structural core

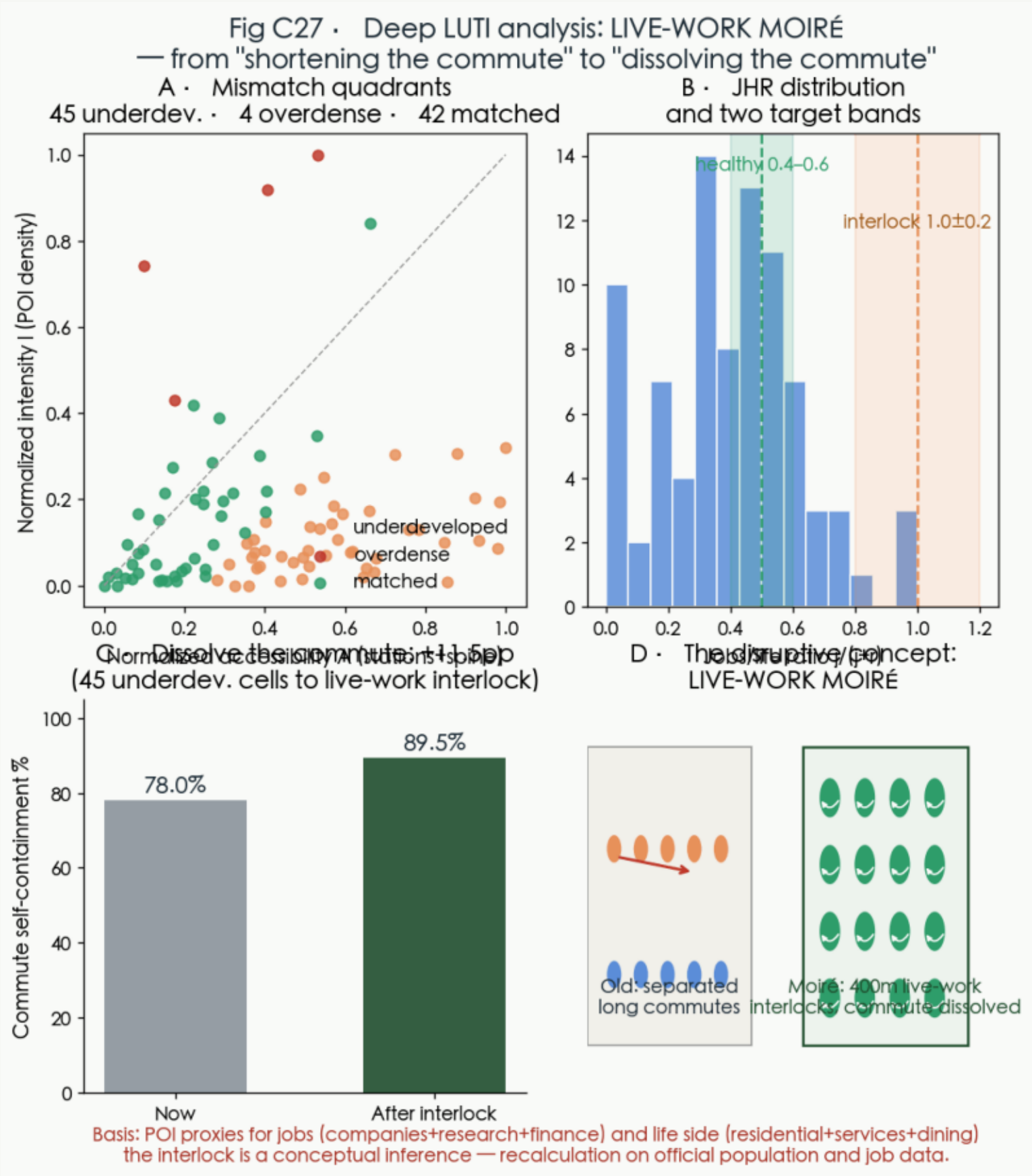
▪ Dazhongsi 20.2%: syntactic periphery, vitality must come from function

▪ After stitching: Dazhongsi +70%, Beijing North +49%, Wudaokou +43%

5,720-node network, computed: low intelligibility is normal for organic fabric

⑤ Deep LUTI: LIVE-WORK MOIRÉ — dissolve the commute

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▪ Disruptive concept: 45 underdeveloped cells rebuilt as 1.0 ± 0.2 live-work interlocks

▪ Commute self-containment 78.0% → 89.5% (+11.5pp, computed)

▪ Mismatch quadrants 45/4/42: where it's wrong → after stitching → what remains

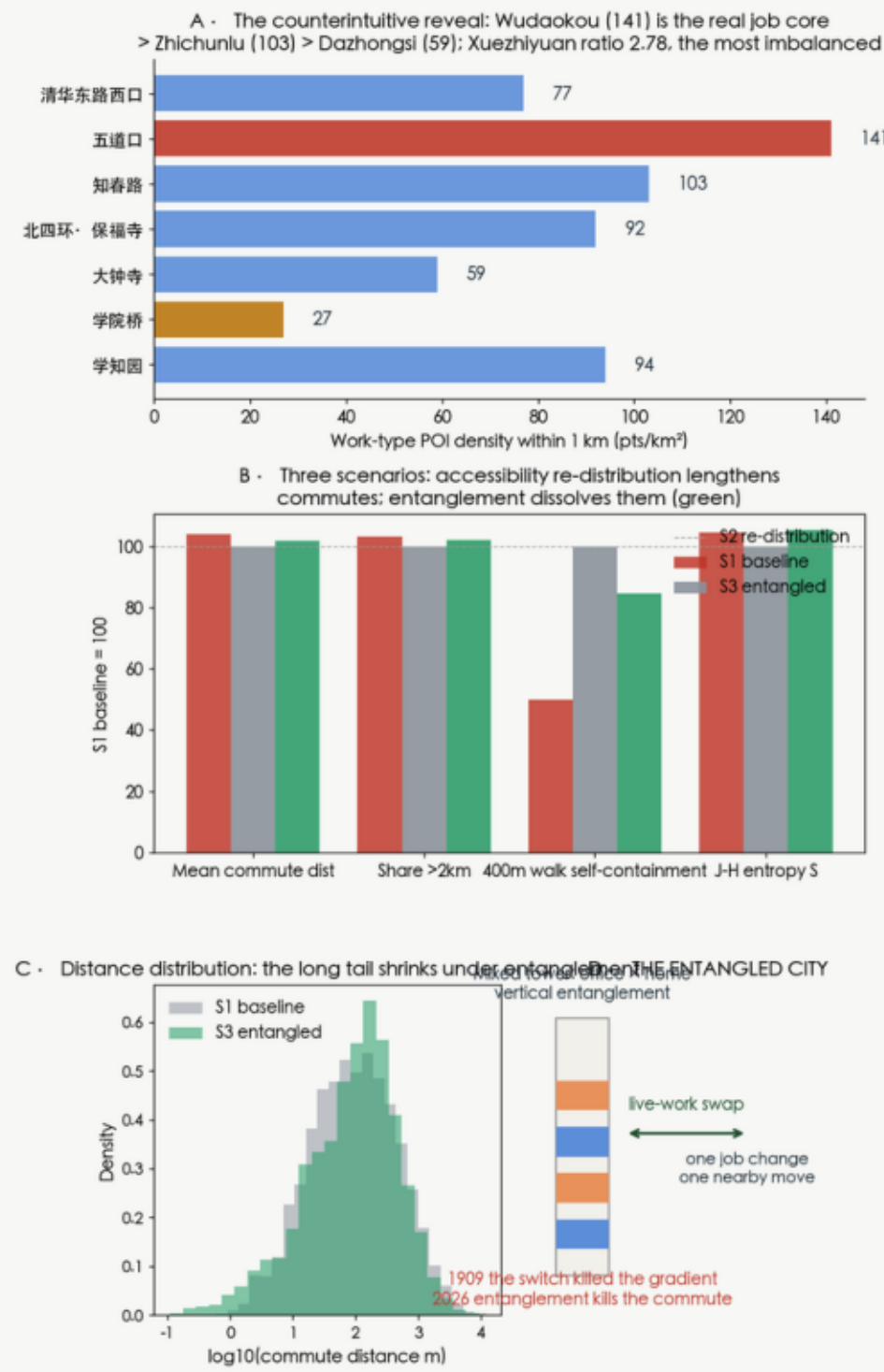
▪ Not a city with fewer commutes — a city that does not need commuting

POI proxies: recalculation on official population and job data

⑥ Three-layer LUTI scenarios: concentration lengthens, entanglement dissolves

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Fig C28 · Three-layer LUTI scenario runs: only live-work entanglement dissolves the commute (Batty-style scenarios)



▪ The reveal: Wudaokou 141/km² is the real job core; Xuezhijuan ratio 2.78, most imbalanced

▪ Three scenarios (gravity OD): concentration +3.9% distance, walking self-containment 7.0%→3.5%

▪ Entangled: entropy S +5.3% (more mixed); rules = paired infill + mixed towers + S in the annual check-up

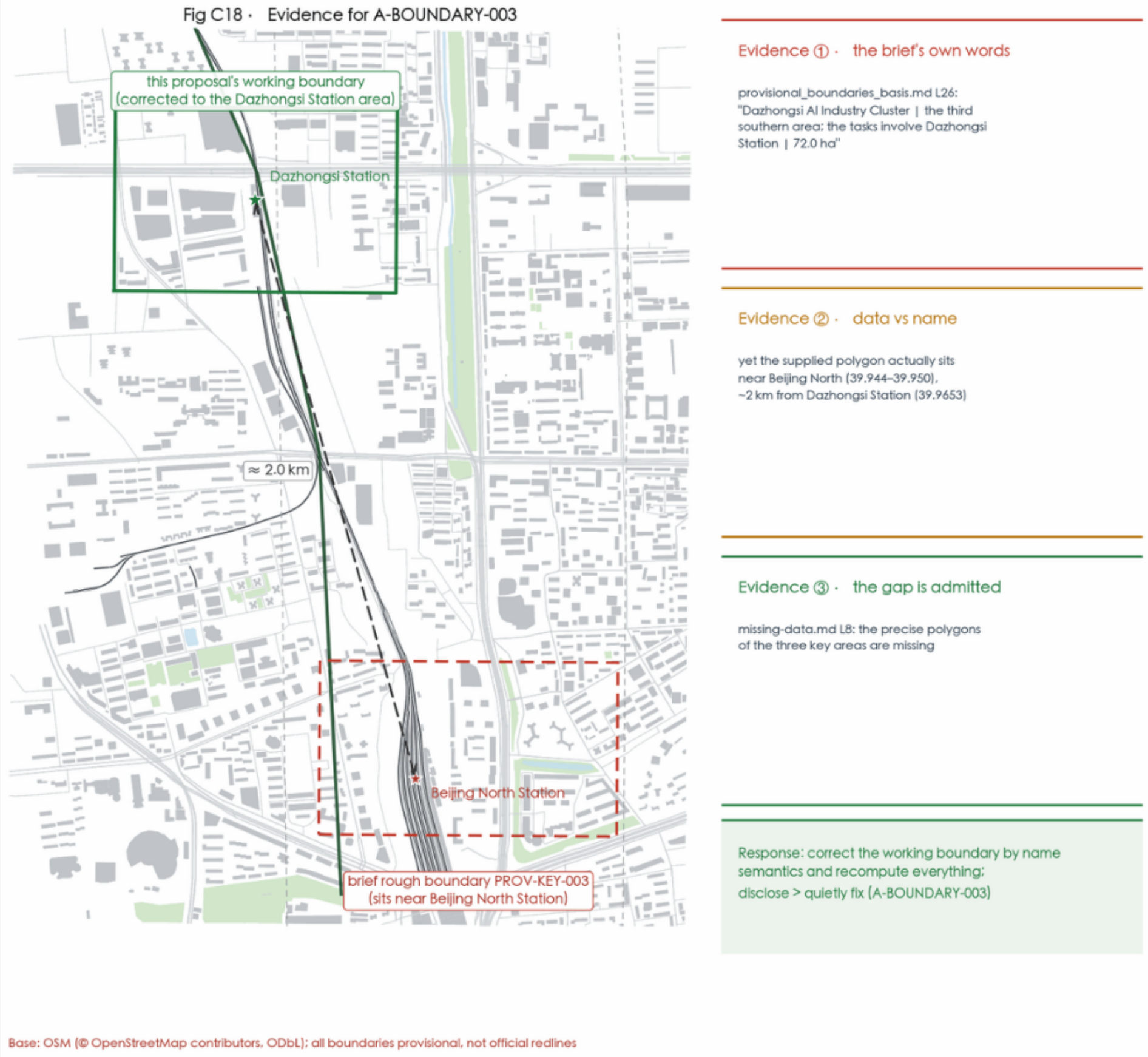
▪ 1909 the switch killed the gradient, 2026 entanglement kills the commuting tide

Gravity OD + POI proxies; recalculation on official OD; floorspace ledger = the brain's first new database

Basis: production-constrained gravity OD ($\beta=1/2500\text{m}$, same basis as λ); POI proxies for jobs/housing; S3 is a conceptual scenario — recalculation on official commute OD (signalling/card data).



⑦ Boundary correction: fixing the brief's rough boundary



- The brief's PROV-KEY-003 rough boundary sits near Beijing North (39.944–39.950)
- Its own basis table names Dazhongsi Station (39.9653) — ~2 km away
- missing-data.md: precise polygons are missing
- Corrected by name semantics, assumption A-BOUNDARY-003, full recalculation on official data

Integrity first: a data mismatch must be disclosed, never quietly fixed

02

Spatial proposal: one line, three folds, two wings, seven nodes

Master plan · Key-area plans & sections · Character & height · Dazhongsi quadrants

⑤ Master plan (real fabric + concept overlay)

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Fig.1 · Master Plan: one line, three folds, two wings, seven nodes (concept)



▪ Light grey = OSM building fabric; white-cased dark = real Jing-Zhang railway

▪ Dark green = Hyper Line spine; red dots = seven stitches; stars = three landmarks

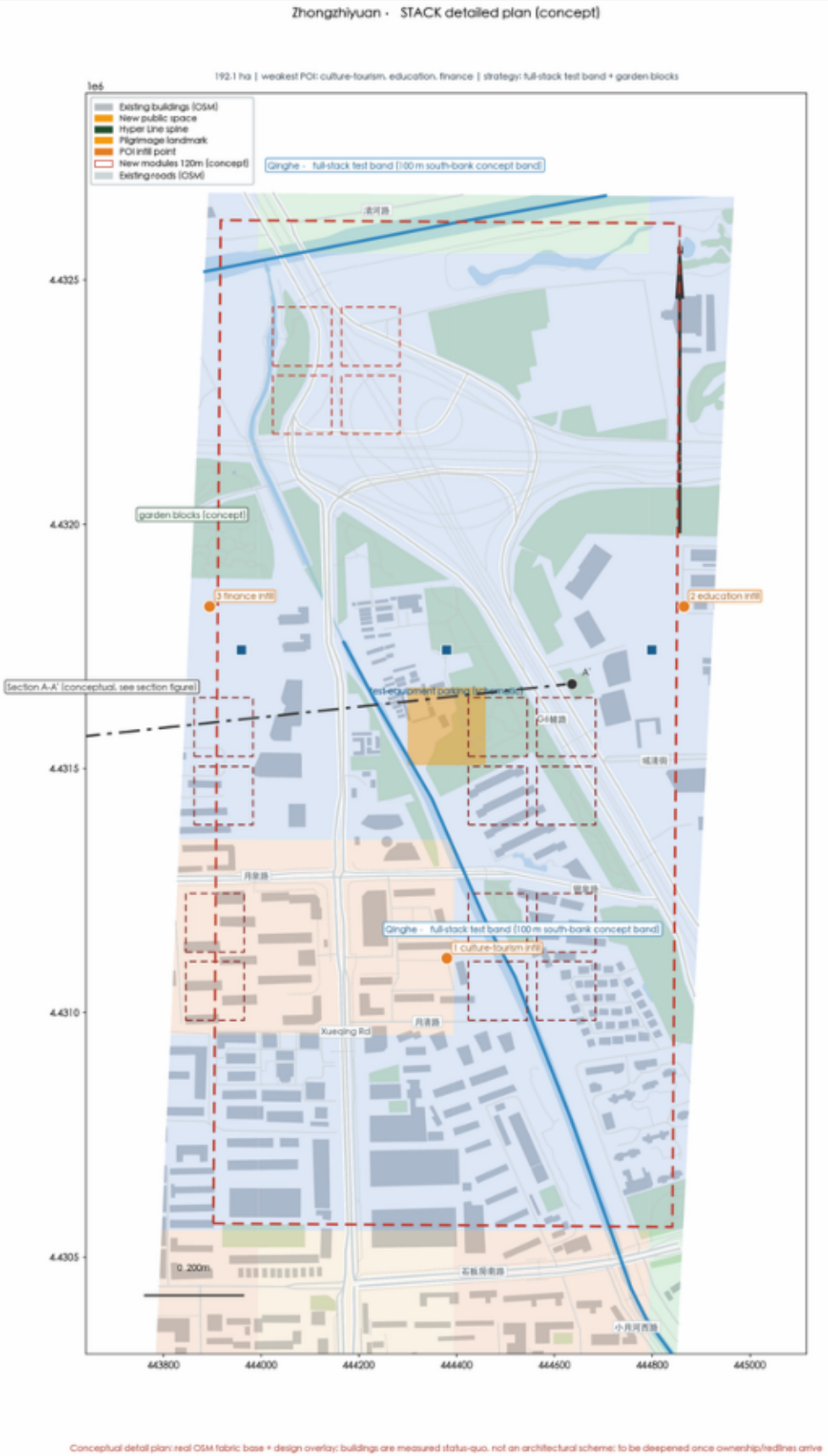
▪ Three coloured frames = three gates: STACK / ORIGIN / FRONT

▪ The base is the real city; only the overlay is design

Base: OSM (@ OpenStreetMap contributors)

⑥ Zhongzhiyuan · STACK (192.1 ha)

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▪ Qinghe full-stack test band: bounded tests in garden blocks

▪ Weakest: culture-tourism/education/finance → reasons to stay

▪ Section A-A': 3.2m footpath + 2.8m cycle band + test-equipment slots

▪ Long term (5–10y): garden-district renewal

⑦ Origin · ORIGIN (104.3 ha)

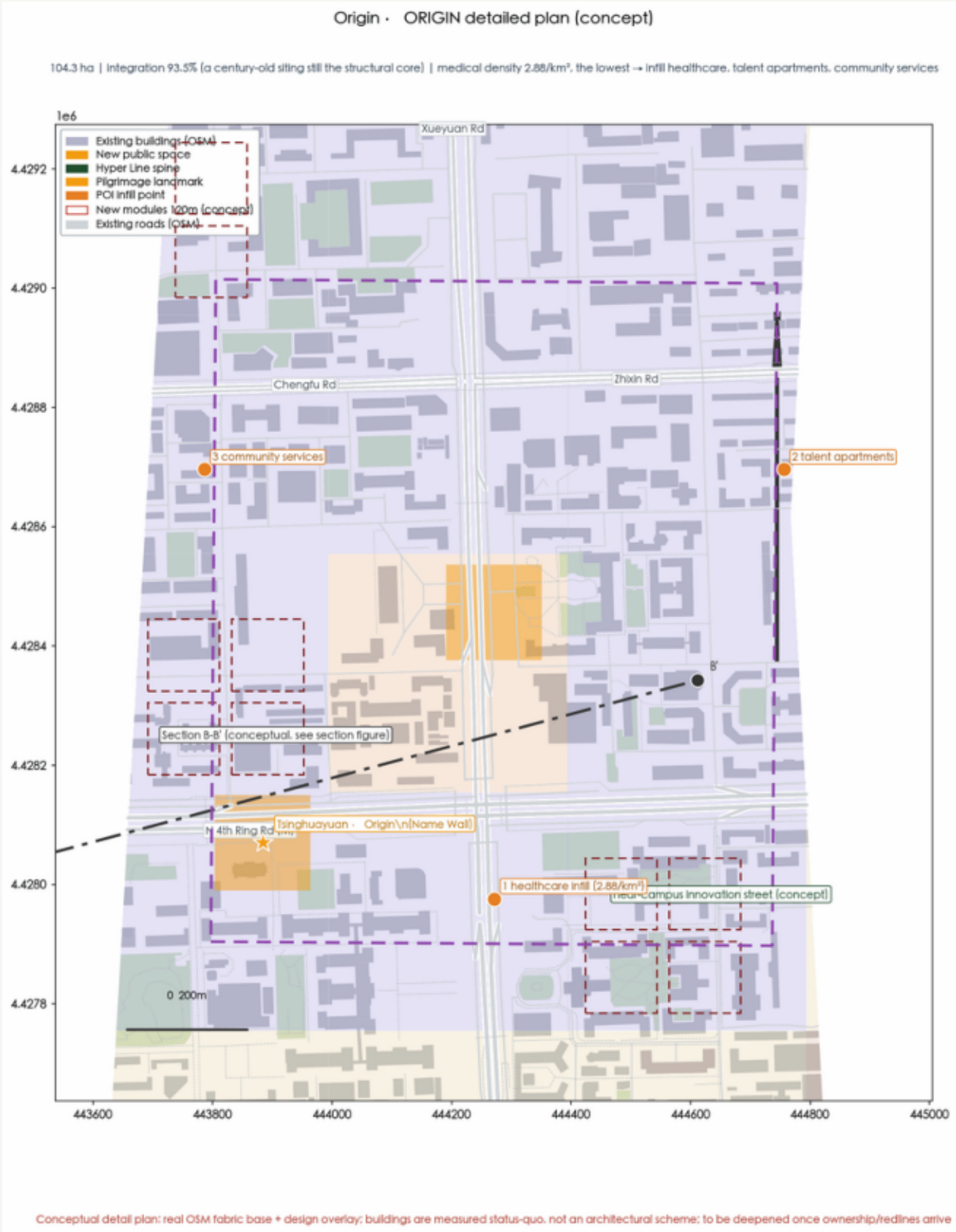
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▪ Tsinghuayuan · Origin Plaza + Name Wall: integration 93.5%

▪ Wudaokou · Phase Plaza: the λ gauge — the city's thermometer

▪ Medical 2.88/km², the lowest → infill healthcare/apartments/services

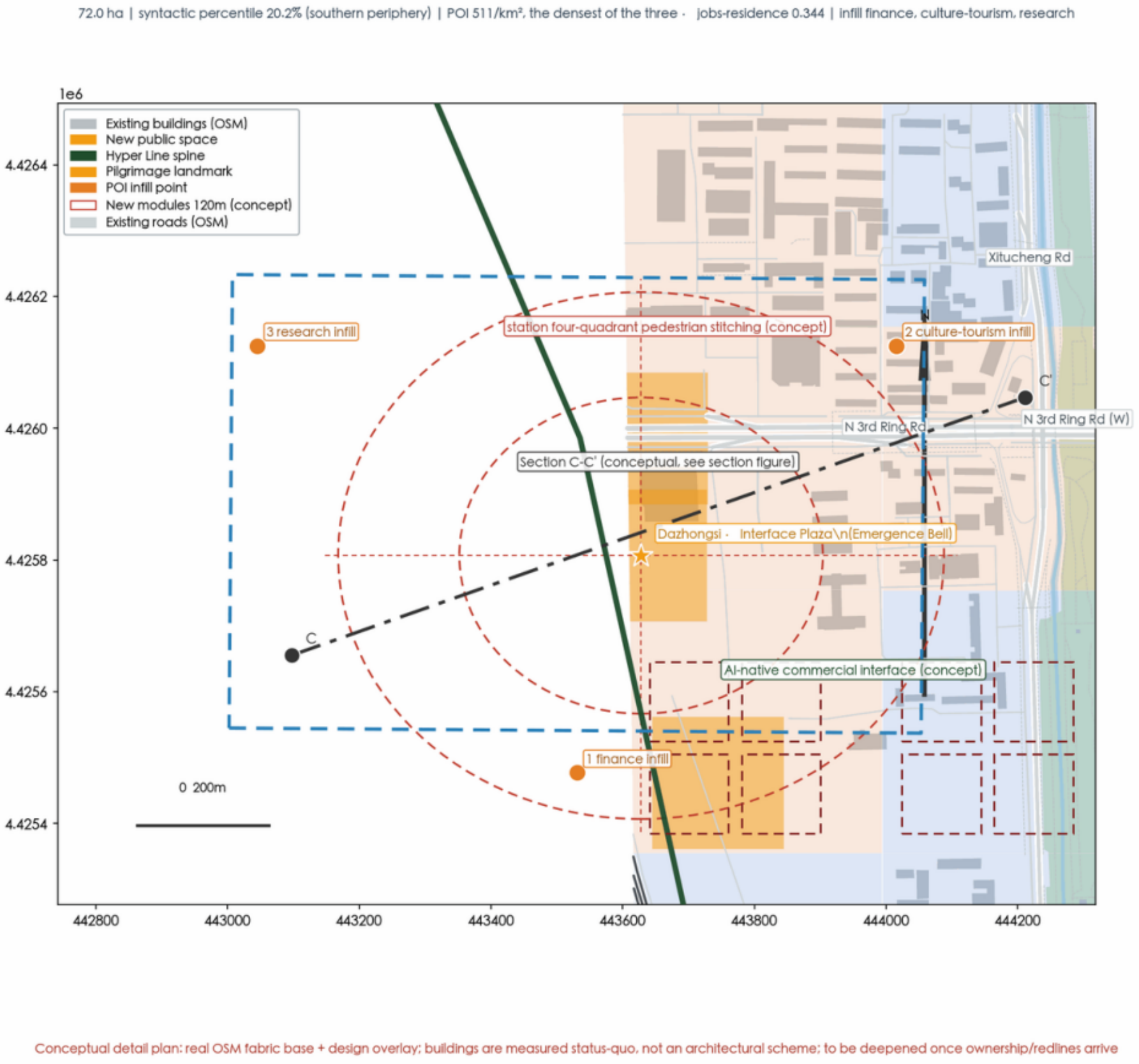
▪ Mid term (3–5y): near-campus innovation street



⑧ Dazhongsi · FRONT (72.0 ha)

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Dazhongsi · FRONT detailed plan (concept)



▪ Syntactic periphery 20.2%, yet the densest POIs at 511/km²

▪ Jobs/life 0.344: life-service-led → infill finance/culture-tourism/research

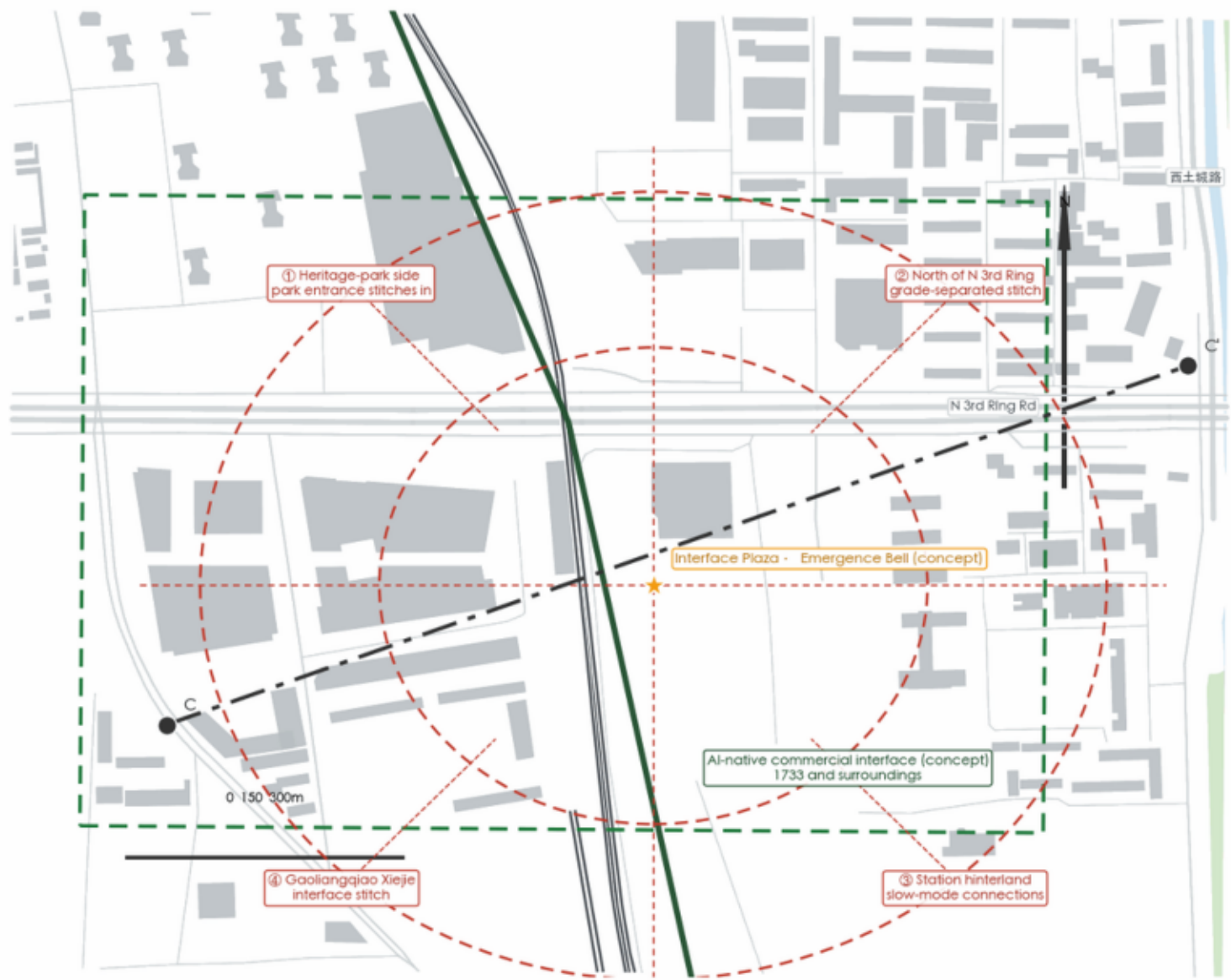
▪ Light intervention, heavy interface: stitch four quadrants; nights for classes & merchants

▪ Near term (1–3y): spine stitching + Dazhongsi interface

⑨ Dazhongsi four-quadrant stitching (enlarged)

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Fig C19 · Dazhongsi station-district four-quadrant stitching, enlarged (conceptual design)



Conceptual schematic: the stitch modes (grade-separation / slow-mode links) are conceptual suggestions, to be deepened once transport, municipal, and ownership data arrive; section C-C' shown in the detailed section figure.

Base: OSM (© OpenStreetMap contributors, ODbL)

- ① park entrance stitches in ② grade-separated stitch north of N3rd Ring
- ③ station-hinterland slow modes ④ Gaoliangqiao Xiejie interface
- Interface Plaza · Emergence Bell + the 1733 AI-native interface
- Section line C-C' matches the detailed plan

⑩ Street sections of the three areas (conceptual · dimension chain · plan index)

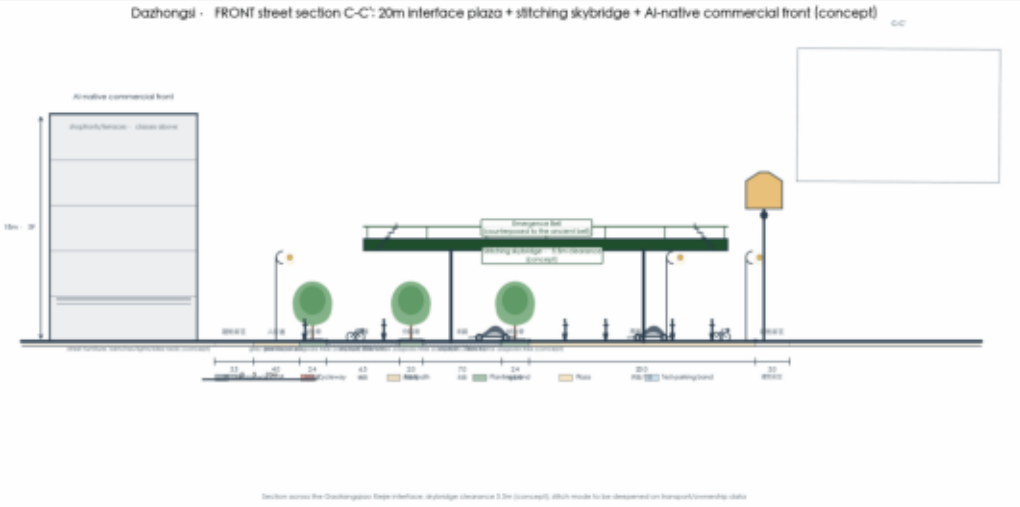
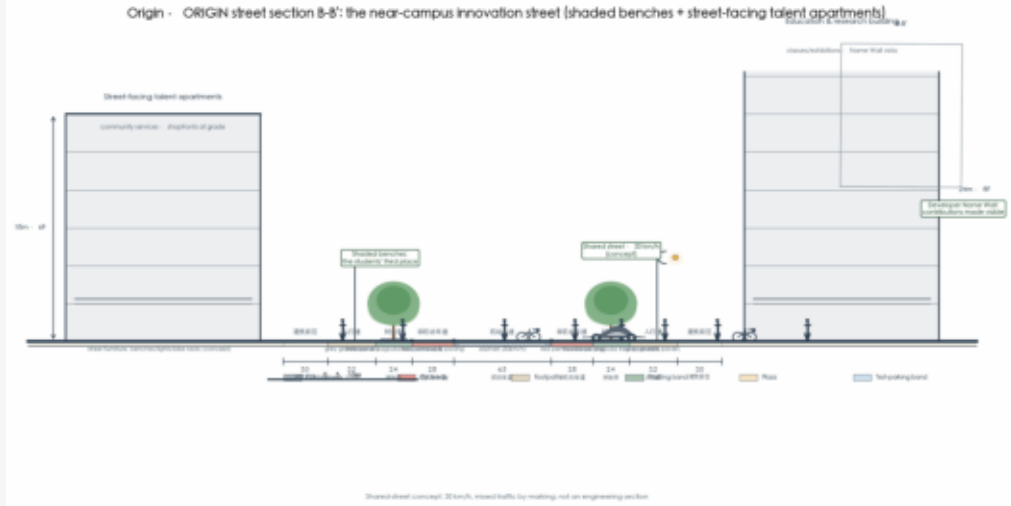
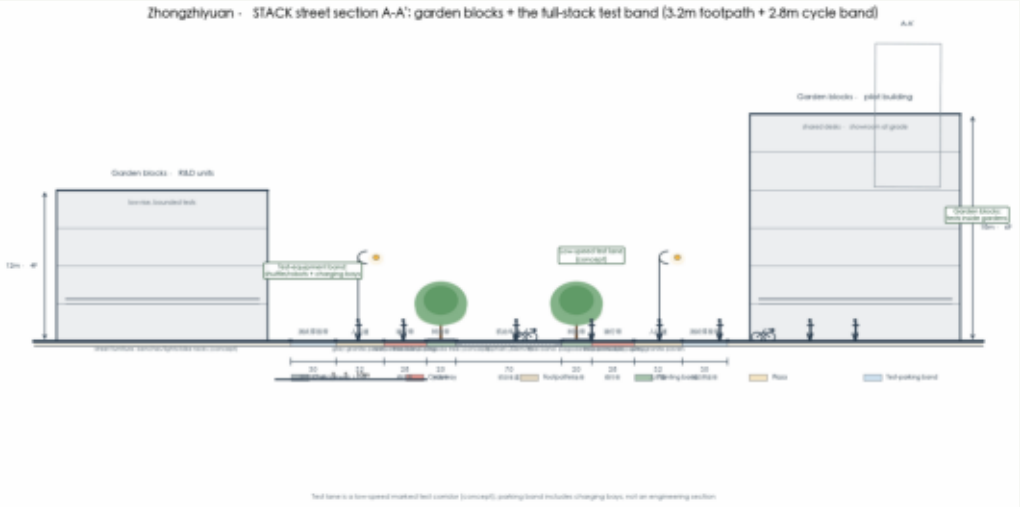
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▪ Each section: building-to-building full section + dimension chain + floors/heights

▪ Scale references: 1.8m people · bikes · cars · low-speed shuttle · trees & lamps

▪ Dazhongsi C-C': 20m interface plaza + stitching skybridge (5.5m) + Emergence Bell

▪ Conceptual sections (not engineering); dimensions pending redline & heritage data

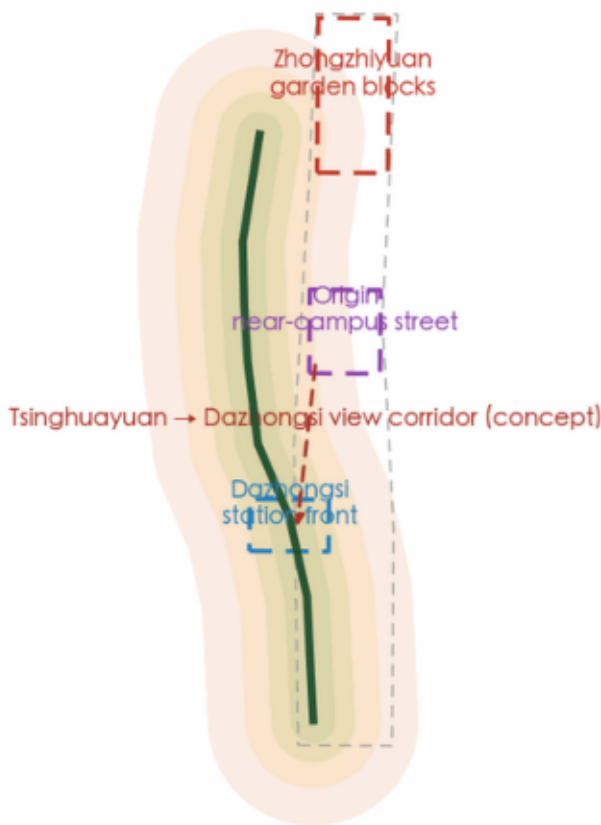


Conceptual sections: design intent; to be deepened on redline, heritage and vertical data

Character & conceptual height zoning

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Conceptual height zoning (schematic):
bands step down toward the spine, low front / high back



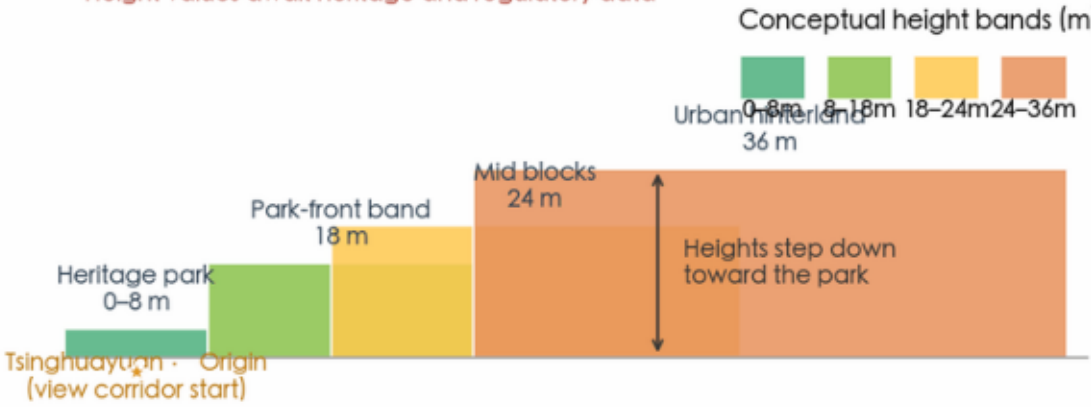
▪ Height bands step down along the spine: 0–8 / 18 / 24 / 36 m

▪ Three rules: garden blocks / near-campus street / station front

▪ Protective overlay (heritage + view corridors) + generative rules (low front, high back)

▪ Height values await heritage & regulatory data (not a regulatory drawing)

Typical section: Origin Community (conceptual, not an engineering section)
Height values await heritage and regulatory data



Schematic: height and character rules are design intent,
to be rechecked once heritage, regulatory, and ownership data arrive
(not a regulatory drawing)

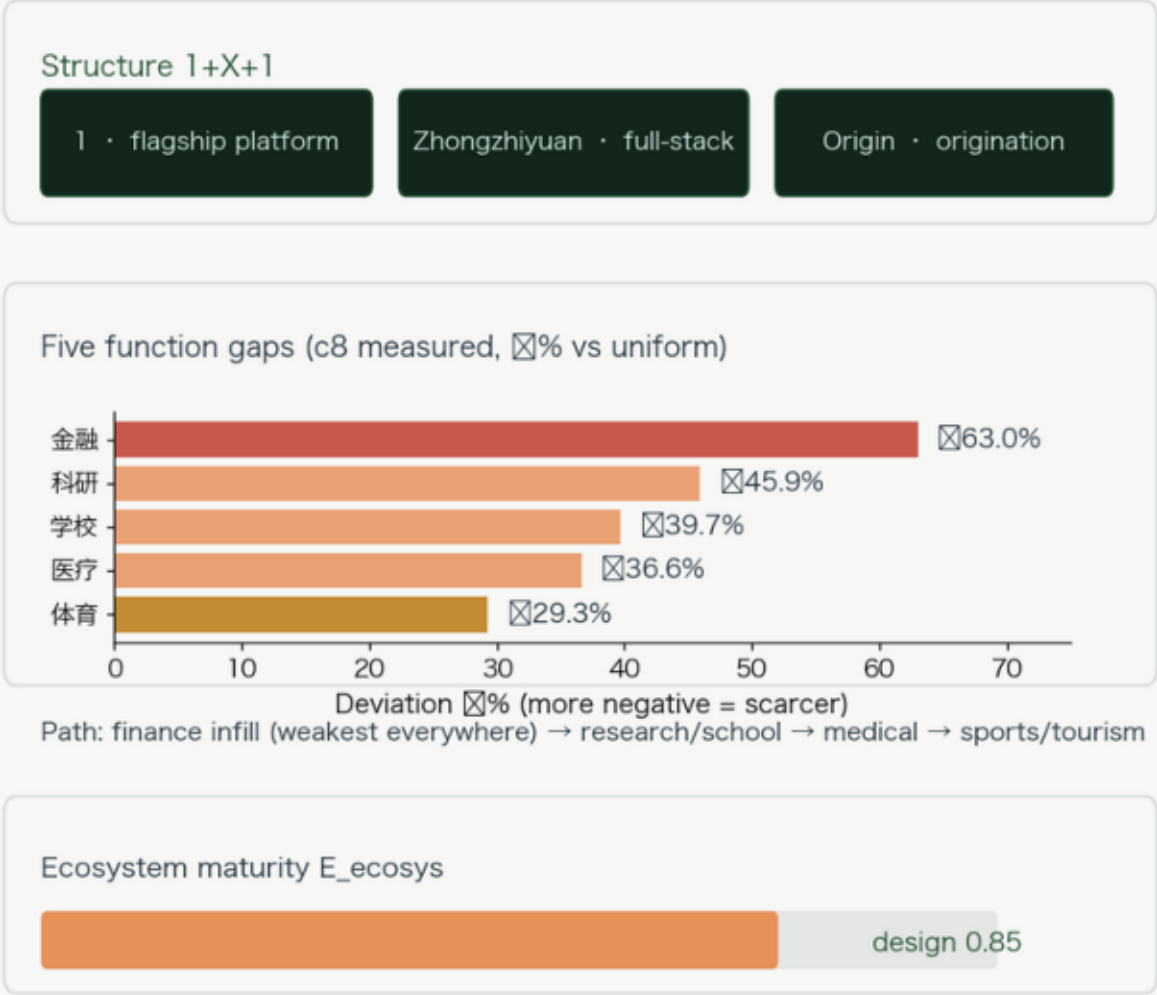
03

Scenarios & ecosystem: the AI ecology and pilgrimage

Ecosystem map & seven cases · Ten scenario cards, five personas · Three pilgrimage landmarks · Operations & timeline

Fig · Industry ecosystem map: the 1+X+1 structure and five function gaps

Fill what is missing — finance is the weakest everywhere; the first ecosystem target (c8/c34 measured)



- Upstream complete: 1,000+ scientists, 13k developers, 180 OPCs, 32 chips
- Middle layer unmeasured: affordability/conversion/compute → annualized
- Seven cases, one transferable mechanism each (King's Cross / Kendall / Cambridge...)
- Five-party six-force conversion + the annual Phase-Forum report as interface

Basis: deviation = (actual/uniform %1); E_{ecosys} = gap closure over five deficit categories (target %20%, provisional); diagnostics only, attraction is conceptual.

Fig · Operations timeline: stitch first, then infill and unlock; three phases with annual gates

A 90-day pilot first; each phase ends with the annual check-up ($\lambda \geq 40\%$ · $H \geq 1.35$); no pass, no next phase

●

Y1 - 3 · Stitching

Seven stitch connections + Dazhongsi front renewal; the 90-day pilot first (fence→paths→meter→operations)

●

Y3 - 5 · Infill

Origin renewal: talent apartments delivered + finance/medical/research infill; live-work pairs (1:1)

●

Y5 - 10 · Unlock

Zhongzhiyuan garden blocks & full-stack test band: 42 CA cells re-allocated; Emergent Brain annualized

Phase gates: percolation 0.904 ☒ · syntax +36% ☒ · entropy +5.3% ☒ — or stop & review; money follows verification

Basis: phasing is conceptual (re-sequenced on official data); gates share sources with the three gates (c2/c4b/c28 measured).

▪ Annual: dev festival / Phase Forum (λ & H report) / open test seasons

▪ Attraction funnel: OPC contest → order plaza → desks → incubator → conversion

▪ Two acceptance gates: $\lambda \geq 40\%$, $H \geq 1.35$ — or stop and review

▪ Phases: 1–3 stitching+interface / 3–5 Origin / 5–10 Zhongzhiyuan

International benchmark matrix: cases × mechanisms × responses × numbers

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Fig C26 · International benchmark matrix: seven global cases, one transferable mechanism, one of our measured numbers				
Benchmark discipline: no copied images or geometry — mechanisms only; each mechanism lands on one of our numbers				
Case	Mechanism		Our response	Measured
London · King's Cross	open the paths and squares first, let the plots grow themselves	stitch the seven nodes first, then the plots		integration +43%~+70%
Boston · Kendall Square	the near-campus conversion circle: density within 1 km of MIT	Origin near-campus street: catching those leaving school		Tsinghuayuan integration 93.5%
Stanford Research Park	campus origination: low-density garden research neighbourhoodzhongzhiyuan garden blocks + the full-stack test band			13,000 developers · 32 chips
Shenzhen · Yuehai Street	the fabric density of high-density startup blocks	Dazhongsi front: a 511 POI/km² station interface		25,476 POIs measured
Tokyo · Shibuya	the renewal paradigm of a content-consumption interface	AI-native commercial front + Emergence Bell (night classes/exhibitions)		10 scenario cards landed
Cambridge Cluster	the symbiotic structure of a university town and an industry belt	the five-party six-force conversion + the annual Phase-Forum report		180 OPCs · 18 vendors adapted
Tel Aviv · Rothschild Blvd	the public life of a startup boulevard	the dev festival + honor wall + spine public space		equity loops 5/5 online
The opportunity: seven mechanisms all hold on one belt — the unique opportunity of Centennial Jing-Zhang: the world's first AI belt designed by urban science and run recursively.				

Seven global cases, one transferable mechanism each (no copied images or geometry)

King's Cross × stitch-first / Kendall × conversion circle / Stanford × garden neighbourhoods

Yuehai × dense fabric / Shibuya × consumption / Cambridge × symbiosis / Rothschild × public life

The opportunity: seven mechanisms hold at once on one belt — Jing-Zhang's unique window

Benchmark discipline: mechanisms transferred, numbers computed locally

04

System & finance: the Emergent Brain and the investment ledger

Emergent Brain architecture · Phase Control Room · Investment & return · Risk & compliance

The Emergent Brain: the city as a recursive program

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- Three layers: sense → compute → decide, with humans always holding the veto
- Four terminals: citizens / enterprises / government / campus
- The recursive loop is the annual check-up: read→compute→tune→verify→re-read
- Red lines: no privacy violation, no over-monitoring, human-reviewable

Phase Control Room: the belt's annual check-up

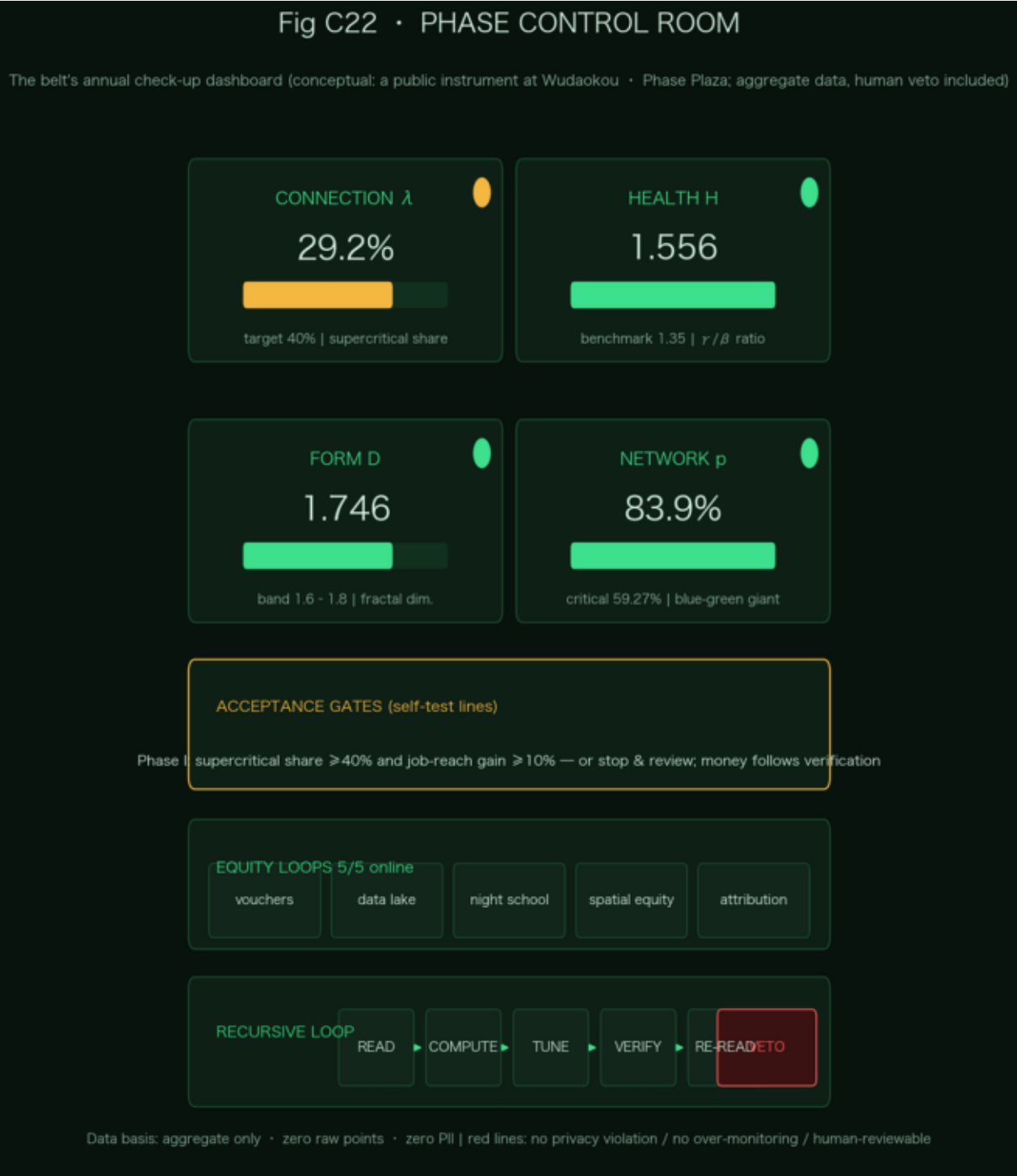
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Four gauges: λ 29.2%→40% target / H 1.556 / D 1.746 / p 83.9%

Live panels: job reach 323→375, match 46.2%→61.5%, η +60%

Physically at Wudaokou · Phase Plaza: a public instrument, not surveillance

Aggregate only, zero raw points, zero PII, human veto



Investment & return (conceptual magnitude)

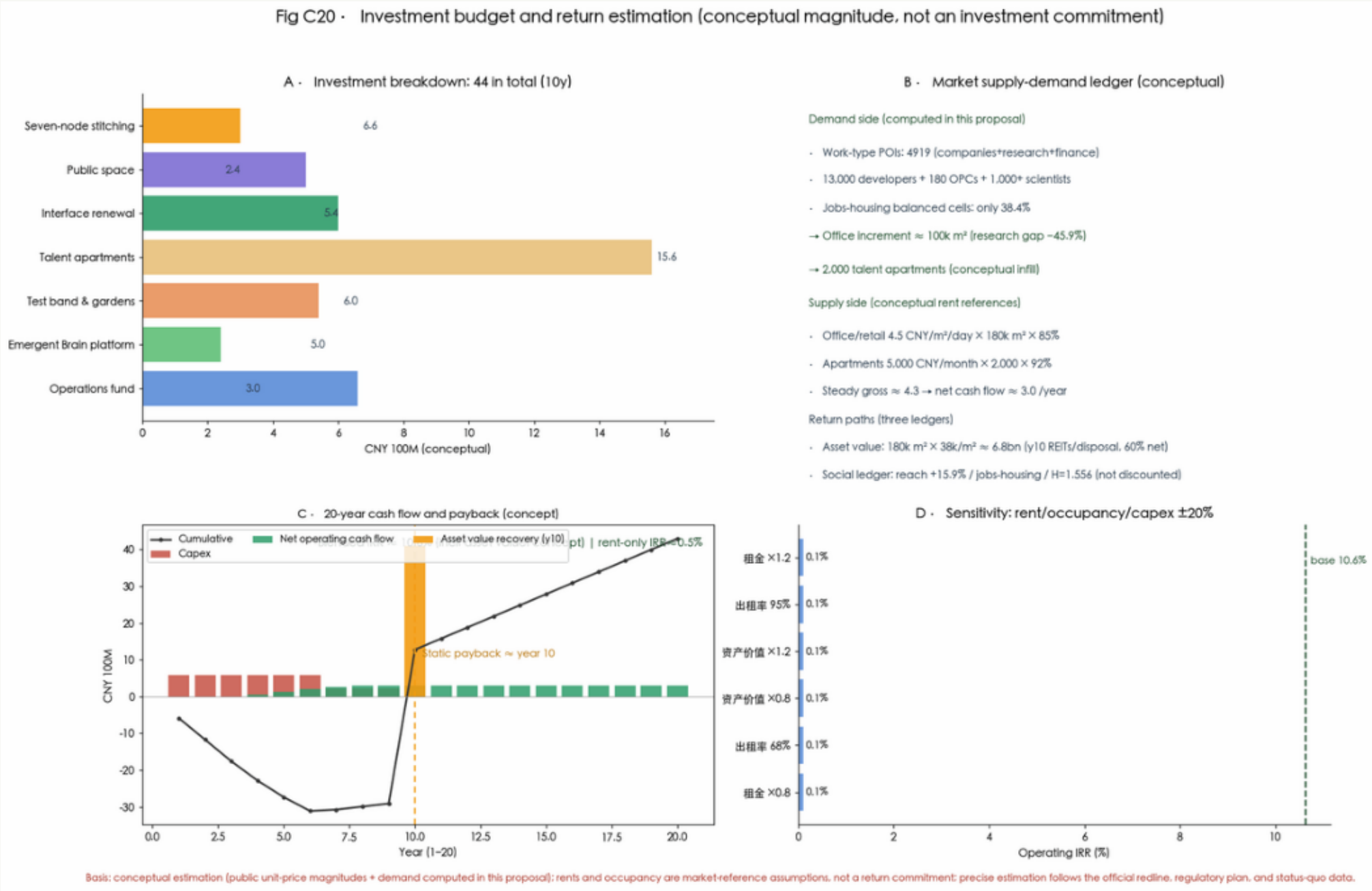
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CNY 4.40bn: stitching 0.66 / space 0.24 / interface 0.54 / apartments 1.56 / test band 0.60 / brain 0.50 / ops 0.30

Three ledgers: rent (IRR≈0.5% alone) + asset (y10 REITs/disposal, blended IRR≈10.6%) + social

Demand: 4,919 work POIs → 100k m² offices; 38.4% balance → 2,000 apartments

Public ledger ≈1.7bn seeks no financial return; market ledger ≈2.7bn runs rent+asset



Conceptual; not an investment commitment; rents & occupancy are replaceable assumptions



05

Conclusion: four sentences, each with numbers

Conclusion dashboard · 1909 switch → 2026 belt

Fig C23 · Design-conclusion dashboard

Four sentences, each with numbers



The future cannot be predicted, but it can be invented. — Michael Batty

Four conclusions share sources with the three gates: c2 percolation / c4b syntax / c28 entropy / c33 capacity — all computed locally

▪ ① Re-grow, not rebuild: D=1.746 / H=1.556 / giant 83.9%

▪ ② Move aside what blocks: 7 gaps→7 stitches / 42 cells / three annualized

▪ ③ Falsifiable: jobs +15.9% / match +15.3pct / η +60%

▪ ④ An OS, not a blueprint: four rulers + brain + check-up + exit

Move aside what blocks emergence, and leave the rest for the city to grow itself.

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This proposal is an open co-creation suggestion; it does not replace formal planning and constitutes no government-reviewed conclusion; provisional boundaries are dashed only, statutory indicators pending.